

1922 C Closest cities

PROBLEM LOGIC & NOTES

→ city numbered from 1 to n

→ from city x , 2 choices

- $|a_x - a_y|$ in that much coin, go from x to y
- OR go to closest city in 1 coins, no matter what the distance b/w them is

→ for a query $x \rightarrow y$ need to return min coin to travel from $x \rightarrow y$

→ we will maintain and precompute 2 things:

- `right[i]` → min cost to go from i to $i+1$
- `left[i]` → min cost to go from i to $i-1$

→ need both because it's possible for Eg:

- need to go $4 \rightarrow 5$
- but 3 is closer than 5 is to 4
- So $4 \rightarrow 3$ (1 coin)
- $4 \rightarrow 5$ ($arr[5] - arr[4]$) coin

→ once done:

- **prefixright** and **prefixleft** to compute cost of moving from city 0 to city i
- OR **prefix left** for moving from city i to city 0

C++ IMPLEMENTATION

```
#include<bits/stdc++.h>
using namespace std;

int main(){

    int t;
    cin>>t;

    while(t--){
        int n;
        cin>>n;

        vector<long long> arr(n);

        for(int i=0;i<n;i++){
            cin>>arr[i];
        }

        vector<long long> right(n,0);
        vector<long long> left(n,0);

        right[0]=1;
        for(int i=1;i<n-1;i++){
            int ldist=arr[i]-arr[i-1];
            int rdist=arr[i+1]-arr[i];

            if(rdist<ldist){
                right[i]=1;
            }
            else{
                right[i]=rdist;
            }
        }

        left[n-1]=1;
        for(int i=1;i<n-1;i++){
            int ldist=arr[i]-arr[i-1];
```

```

        int rdist=arr[i+1]-arr[i];

        if(ldist<rdist){
            left[i]=1;
        }
        else{
            left[i]=ldist;
        }
    }

    vector<long long> prefr(n,0);
    for(int i=1;i<n;i++){
        prefr[i]=prefr[i-1]+right[i-1];
    }

    vector<long long> prefl(n,0);
    for(int i=1;i<n;i++){
        prefl[i]=prefl[i-1]+left[i];
    }

    int m;
    cin>>m;

    while(m--){
        int x,y;
        cin>>x>>y;

        x--;
        y--;

        if(x>y){
            cout<<prefl[x]-prefl[y]<<"\n";
        }
        else{
            cout<<prefr[y]-prefr[x]<<"\n";
        }
    }
}

```

```
    return 0;  
}
```